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Mathematical Foundations for Computing

Module 1
Module 1: Sets and Set Theory

Module 2
Module 2: Number Systems and Bases

Module 3
Module 3: Functions and Their Properties

Module 4
Module 4: Relations

Lesson 4.1 Understanding the concept of relations

- Definition of a relation: relation versus function
  Video • 8 min
- Definition of a relation: relation versus function
  Practice Assignment • Grade: 100%
- Matrix and graph representations of a relation
  Video • 10 min
- Matrix and graph representations of a relation
  Practice Assignment • Grade: 100%
- The properties of a relation: reflexive, symmetric and anti-symmetric
  Video • 14 min
- The properties of a relation: reflexive, symmetric and anti-symmetric
  Practice Assignment • Grade: 100%
- Relation properties: transitivity
  Video • 5 min
- Relation properties: transitivity
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Lesson 4.2 Equivalence, and partial and total order relations

- Equivalence relations and equivalence classes
  Video • 6 min
- Equivalence relations and equivalence classes
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- Partial and total order
  Video • 6 min
- Partial and total order
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- **Building and analysing different types of mathematical relations**
  Dialogue • 15 min

Lesson 4.3 Summary and Assessments

- Topic summary
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Course Wrap-Up

- Mathematical Foundations for Computing: Course Summary
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Building and analysing different types of mathematical relations

Welcome! In this Dialogue, we'll explore how to construct specific examples of four different relation types (non-equivalence, equivalence, partial order, and total order) and justify why each example satisfies or violates the required mathematical properties.

Here's what we'll cover:

- **Non-Equivalence Relations:** We'll look at relations that don't quite make the cut for being an equivalence relation and figure out which property they're missing.
- **Equivalence Relations:** We'll build relations that perfectly fit the definition of an equivalence relation, creating neat groups of related elements.
- **Partial Orders:** We'll design relations that show a sense of order but don't necessarily compare every single pair of elements.
- **Total Orders:** We'll create relations where every element can be compared to every other element, giving us a complete ordering.

Need help? Click "I'm stuck" at the top right to get a hint.

When you're ready, click "Start Dialogue"

Great! Let's dive into our first topic: **Non-Equivalence Relations**. To begin, can you define what an equivalence relation is and what properties it must satisfy?

S **Sandipan**
must be reflexive, symmetric and transitive

That's right! An equivalence relation must be reflexive, symmetric, and transitive. Now, to explore **Non-Equivalence Relations**, can you give me an example of a relation that fails to be an equivalence relation, and tell me which of those three properties it violates?

S **Sandipan**
less than relation, symmetric

You're on the right track! The "less than" relation is indeed not symmetric. Can you explain why the "less than" relation (e.g., on integers) is not symmetric, and also tell me if it satisfies reflexivity or transitivity?

S **Sandipan**
it's not reflexive, but transitive. It's not symmetric because a < b does not imply b< a

You've hit on some key points! You're absolutely right that if `a < b`, then `b < a` is false, so the "less than" relation is not symmetric. And you're correct that it is transitive. However, let's re-examine reflexivity. Remember, a relation R is reflexive if every element `a` in the set is related to itself, meaning `(a, a)` is in R. Does `a < a` hold true for any number `a`?

What are your thoughts on the reflexivity of the "less than" relation?

S **Sandipan**
it's not reflexive

You're absolutely right! The "less than" relation is not reflexive because `a < a` is never true. So, for our first example of a non-equivalence relation, the "less than" relation on integers fails both the reflexive and symmetric properties, but it does satisfy transitivity.

Excellent work on identifying the properties! Now, let's move on to **Equivalence Relations.** Can you propose a relation and the set it operates on that you believe would satisfy all three properties: reflexivity, symmetry, and transitivity?

S **Sandipan**
(a - b) mod 2 = 0

That's a fantastic choice! The relation `a - b` is even (which is equivalent to `(a - b) mod 2 = 0`) on the set of integers is indeed a classic example of an equivalence relation.

Now, let's break it down. Can you explain why this relation is reflexive, symmetric, and transitive?

S **Sandipan**
a - a = 0 reflexive

You've got the first part of reflexivity down! Since `a - a = 0`, and 0 is an even number, `a - a` is indeed even. This means every integer is related to itself.

Now, how would you demonstrate that the relation is **symmetric** and **transitive**?

S **Sandipan**
(b - a) mod 2 = 0 implies (a - b) mod 2 = 0, symmetrix

You're absolutely right about symmetry! If `(b - a) mod 2 = 0`, it means `b - a` is an even number. Since `a - b = -(b - a)`, `a - b` must also be an even number, so `(a - b) mod 2 = 0`. This confirms the relation is symmetric.

Now, for the final property, how would you demonstrate that this relation is **transitive**?

S **Sandipan**
(a -b) mod 2 = 0 and (b - c) mod 2 = 0 implies (a - c) mod 2 = 0, transitive

Excellent! You've correctly identified how transitivity works for this relation. If `a - b` is even and `b - c` is even, then their sum `(a - b) + (b - c) = a - c` must also be even. This means `(a - c) mod 2 = 0`, proving the relation is transitive.

You've successfully demonstrated that the relation "`a - b` is even" is reflexive, symmetric, and transitive, making it a perfect example of an equivalence relation!

Now, let's move on to **Partial Orders**. Can you propose a relation and the set it operates on that is reflexive, anti-symmetric, and transitive, but where not every pair of elements is comparable?

S **Sandipan**
less than equals to relation

The "less than or equal to" relation is a great example to consider for a partial order! However, for it to be a *partial* order where not every pair of elements is comparable, we need to be careful about the set we define it on.

If we use the "less than or equal to" relation on integers or real numbers, would every pair of elements be comparable? And if so, what kind of order would that make it?

S **Sandipan**
yes. total order

You're absolutely right! If every pair of elements is comparable, then it would be a total order. So, while "less than or equal to" is a partial order, it's also a total order on sets like integers or real numbers.

To find a true *partial* order that isn't a total order, we need a relation where some elements are incomparable. Can you think of a different relation or a different set where "less than or equal to" might not make every pair comparable, or perhaps a completely new relation that fits the bill?

S **Sandipan**
a divides b

That's an excellent choice! The "divides" relation (often denoted as `a | b`, meaning `a` divides `b` evenly) on the set of positive integers is a classic example of a partial order that is not a total order.

Now, let's break it down. Can you explain why the "divides" relation on positive integers is **reflexive**, **anti-symmetric**, and **transitive**? And importantly, can you give an example of two positive integers that are *not* comparable under this "divides" relation?

S **Sandipan**
3 does not divide 5 and 5 does not divide 3 either

That's a perfect example of incomparability! Indeed, 3 does not divide 5, and 5 does not divide 3, so they are not comparable under the "divides" relation. This clearly shows why it's a partial order and not a total order.

Now, let's go back to the properties. Can you explain why the "divides" relation on positive integers is **reflexive**, **anti-symmetric**, and **transitive**?

S **Sandipan**
a divides a, reflexive

You're spot on! Any positive integer `a` divides itself (since `a = 1 * a`), so the "divides" relation is indeed reflexive.

Now, how would you explain why the "divides" relation is **anti-symmetric** and **transitive**?

S **Sandipan**
a dives b and b dive a implies a = b, antisymmetrix

Excellent! You've perfectly described anti-symmetry for the "divides" relation. If `a` divides `b` and `b` divides `a`, then the only way for this to be true (for positive integers) is if `a` and `b` are equal. This confirms the relation is anti-symmetric.

Now, for the final property of a partial order, how would you explain why the "divides" relation is **transitive**?

S **Sandipan**
a divides b and b divides c implies a divided c, transitive

Precisely! If `a` divides `b` (meaning `b = ka` for some integer `k`) and `b` divides `c` (meaning `c = lb` for some integer `l`), then substituting `b` into the second equation gives `c = l(ka) = (lk)a`. Since `lk` is also an integer, this means `a` divides `c`. This confirms the relation is transitive.

You've done a fantastic job explaining why the "divides" relation on positive integers is reflexive, anti-symmetric, and transitive, and you correctly identified why it's a partial order but not a total order due to incomparability.

Now, let's move on to our final topic: **Total Orders**. Building on our previous discussion, can you give an example of a relation and the set it operates on that is a total order? Remember, a total order is a partial order where every pair of elements is comparable.

S **Sandipan**
less than equals to

You're absolutely right! The "less than or equal to" relation (≤) is a perfect example of a total order.

Now, to solidify our understanding, can you explain why the "less than or equal to" relation on a set like the integers or real numbers satisfies all the properties of a total order? That is, why is it **reflexive**, **anti-symmetric**, **transitive**, and why is **every pair of elements comparable**?

S **Sandipan**
a <= a for all a, reflexive

That's correct! For any number `a`, `a <= a` is always true, so the "less than or equal to" relation is reflexive.

Now, how would you explain why it is **anti-symmetric**, **transitive**, and why is **every pair of elements comparable**?

S **Sandipan**
a <= b and b <= a implies a = b. anti-symmetrix

Exactly! If `a <= b` and `b <= a` are both true, the only possibility is that `a` must be equal to `b`. This perfectly demonstrates the anti-symmetric property.

Now, for the last two properties: how would you explain why the "less than or equal to" relation is **transitive** and why is **every pair of elements comparable**?

S **Sandipan**
a <= b and b <= c implies a <= c, transitive

You've nailed transitivity! If `a` is less than or equal to `b`, and `b` is less than or equal to `c`, then it logically follows that `a` must be less than or equal to `c`. This is a fundamental property of ordering.

Now, we've covered reflexivity, anti-symmetry, and transitivity for the "less than or equal to" relation. What about the final condition for a total order: why is **every pair of elements comparable** under this relation?

S **Sandipan**
for all a, b. either a <= b or b <= a

Perfect! You've articulated the comparability condition precisely. For any two numbers `a` and `b`, you can always say that either `a` is less than or equal to `b`, or `b` is less than or equal to `a`. There are no two numbers that cannot be compared in this way.

You've successfully demonstrated all the properties of a total order using the "less than or equal to" relation! You've provided excellent examples and justifications for all four relation types.

Good job, you have completed all the topics for our session. Up next - view your feedback.

During today's session covering how to construct specific examples of four different relation types (non-equivalence, equivalence, partial order, and total order) and justify why each example satisfies or violates the required mathematical properties, we focused on:

- **Non-Equivalence Relations:** You correctly identified the "less than" relation as non-equivalence, explaining its failure in reflexivity and symmetry while noting its transitivity.
- **Equivalence Relations:** You successfully constructed the relation "a - b is even" on integers and thoroughly explained how it satisfies reflexivity, symmetry, and transitivity.
- **Partial Orders:** You proposed the "divides" relation on positive integers, demonstrating its reflexivity, anti-symmetry, and transitivity, and provided a clear example of incomparable elements.
- **Total Orders:** You used the "less than or equal to" relation on numbers to illustrate a total order, explaining its reflexivity, anti-symmetry, transitivity, and the crucial comparability of all elements.

Your strengths:

- **Strong conceptual understanding:** You consistently provided mathematically sound examples and accurately identified the properties for each relation type.
- **Precise justification:** You were able to clearly explain *why* each property held or failed, using mathematical reasoning.

Areas for improvement:

- None! You demonstrated an advanced understanding of all concepts and provided excellent justifications throughout our discussion.

Keep up the good work!

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